MANAGEMENT REVIEW

Ames Housing Price Prediction

End-to-end ML pipeline · Detailed analysis for decision

0.129	$29.3k	0.864	80
OOF RMSE (log)	OOF RMSE ($)	OOF R^2	Features used

Data Analytics Team · October 2026

Session roadmap

01 **Executive summary & decision ask**

Results at a glance and what we need from management

02 **Business context & data profile**

Objective, success criteria, sample, missingness

03 **What drives sale price**

Correlations, neighborhood, quality, size, age

04 **Feature pipeline**

Engineering, selection path, final 80 features

05 **Model training & validation**

Algorithms, CV scores, residuals, ensemble logic

06 **Test predictions & artifacts**

Submission distribution, delivery package

07 **Recommendations & next steps**

Deploy, monitor, improve — owners and timing

Production-ready ensemble with ~$29k typical absolute error

0.129

OOF RMSE (log)

$29,250

OOF RMSE (USD)

$15,200

OOF MAE (USD)

0.864

OOF R^2

DECISION ASK

Approve deployment of the 4-model ensemble (HistGradientBoosting + Ridge + RandomForest + GradientBoosting) for batch valuation scoring. Expected typical absolute error $15k–$30k on homes in the $100k–$250k band. Full pipeline, models, and submission file are ready for use.

Strongest drivers

OverallQual, TotalSF (engineered), HouseAge, GarageCars, KitchenQual / ExterQual. Quality and size explain most of the variance.

Method integrity

All imputation, encoding, and selection fit on train only. 212 engineered features reduced to 80 via variance, correlation, and RFE without CV loss.

Known risks

Two extreme GrLivArea outliers; sparse luxury features (pool, tennis). Tree models already dampen their influence. Monitor residual drift post-deploy.

Build a transparent, reproducible automated valuation model

Objective

Predict SalePrice for held-out Ames homes with accuracy suitable for listing guidance and portfolio screening. Output must be explainable and fully auditable.

Success criteria

OOF log-RMSE ≤ 0.14 and absolute RMSE competitive with industry AVMs (roughly 10–18% of median price). Clear feature importance for stakeholders.

Audience & decision

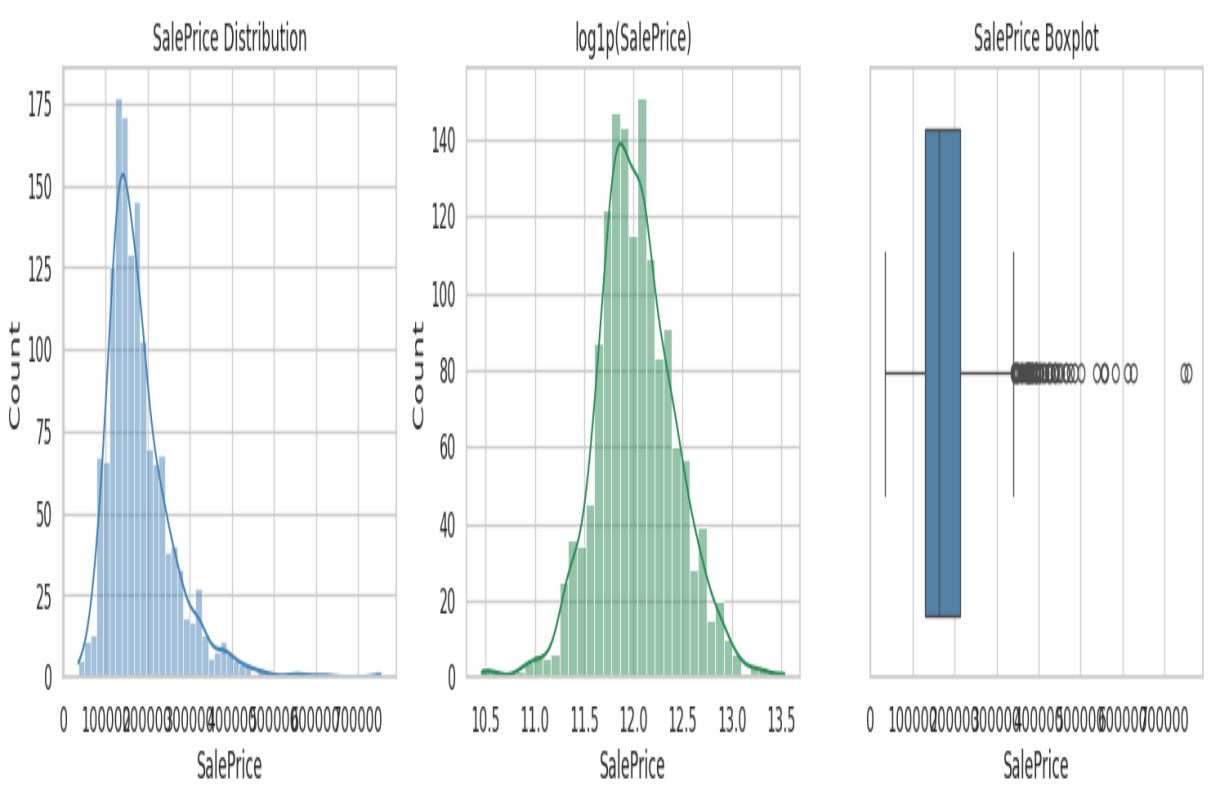
Management go/no-go on internal deployment. Secondary use: optional public leaderboard submission for external benchmarking.

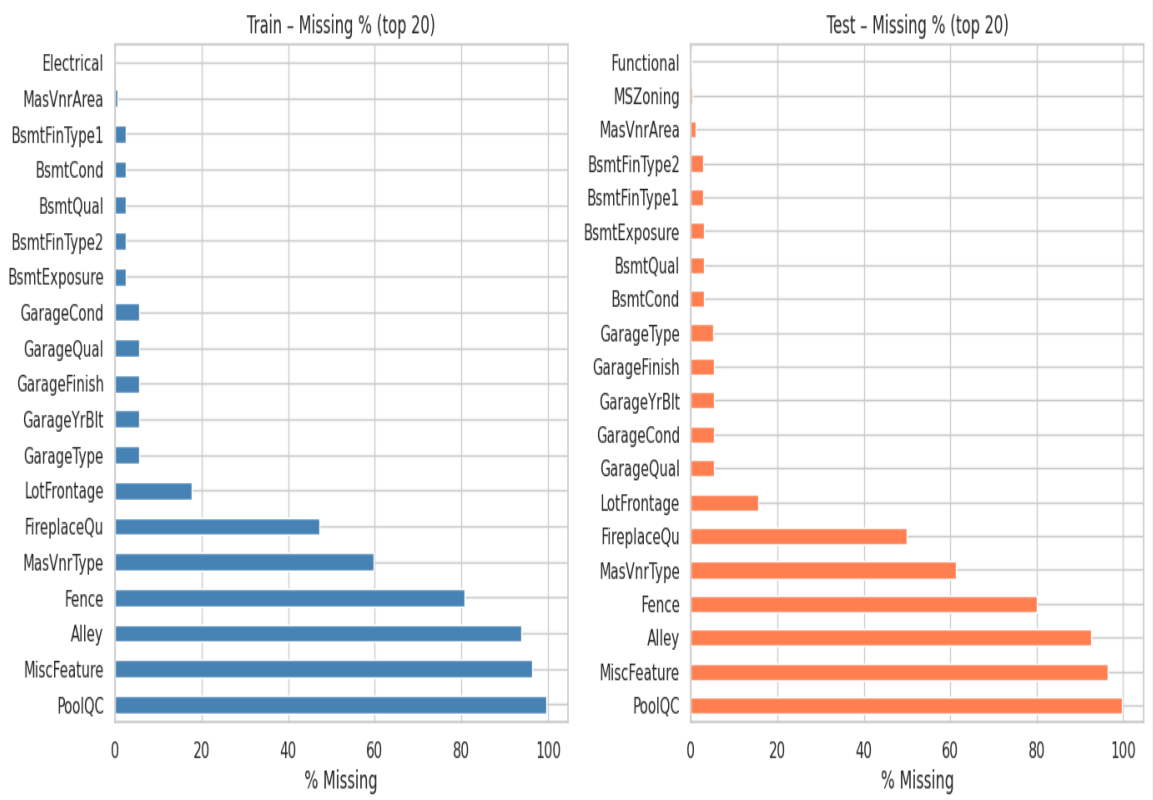
Constraints

No external data sources. Only the provided 2006–2010 Ames train/test files. Strict train-only fitting to prevent leakage. Reproducible notebooks required.

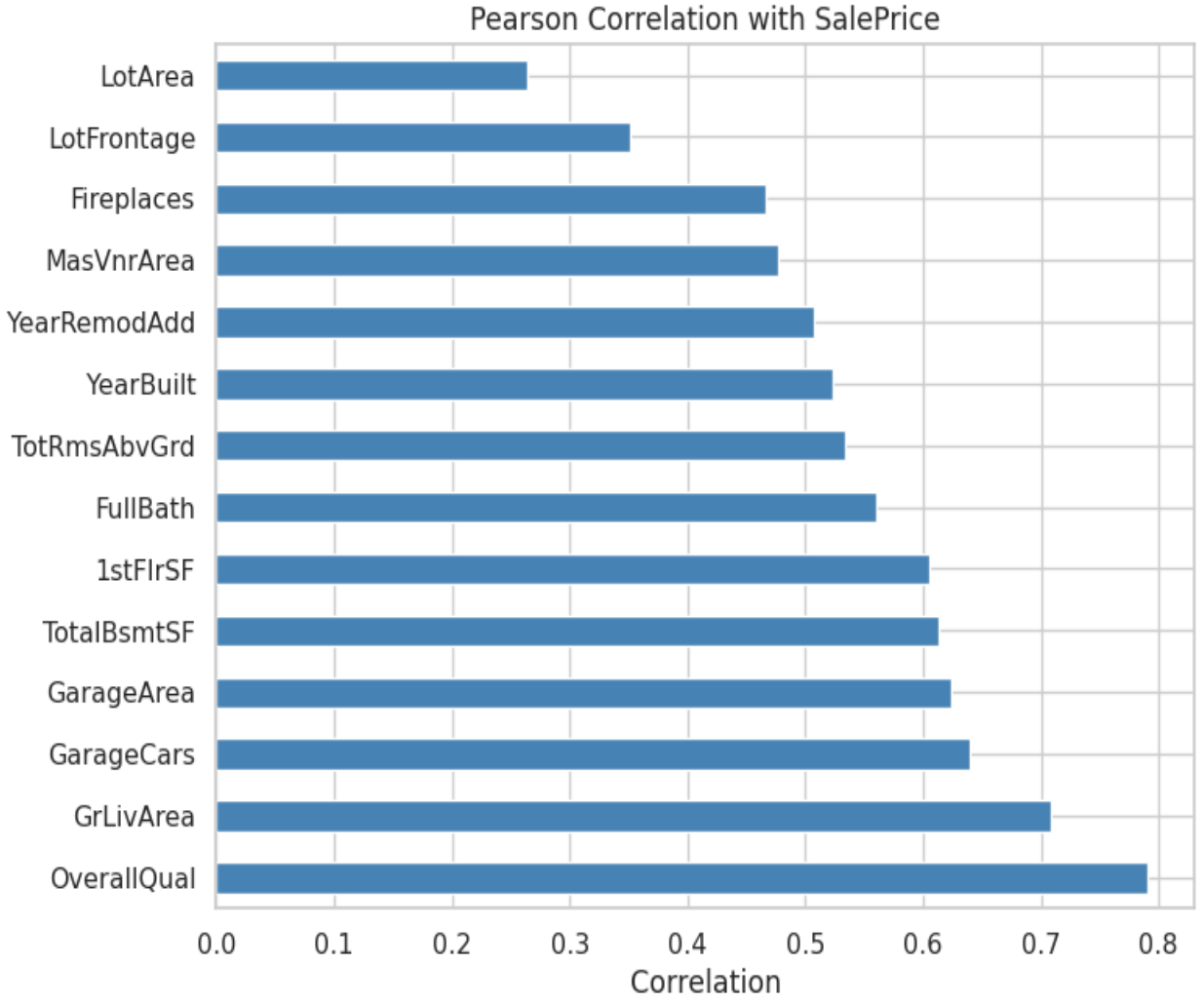
1,460 labeled + 1,459 test homes · informative missingness, not random gaps

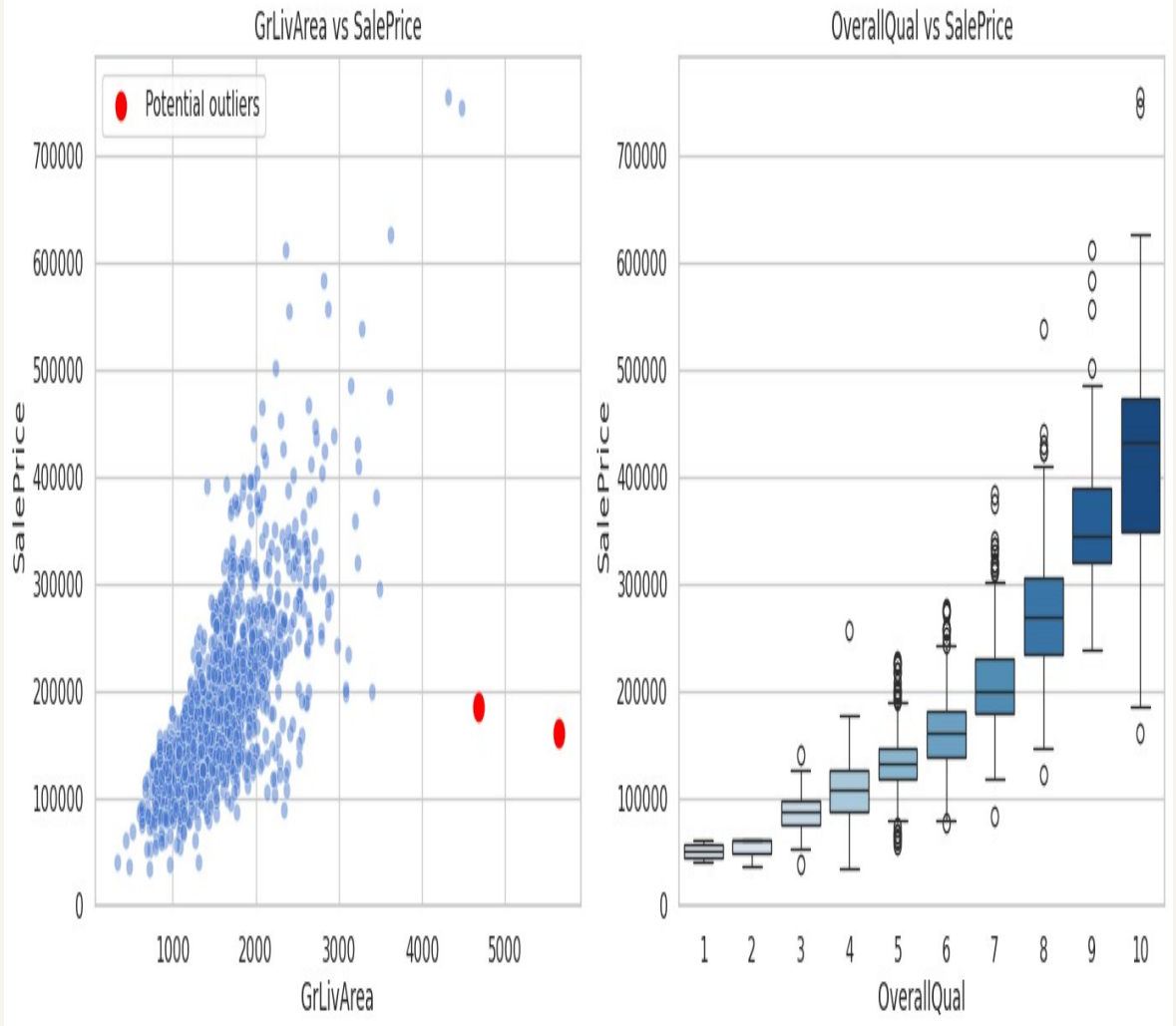
1,460	1,459	81	$163k	$181k	1.88
Train homes	Test homes	Raw columns	Median price	Mean price	Target skew



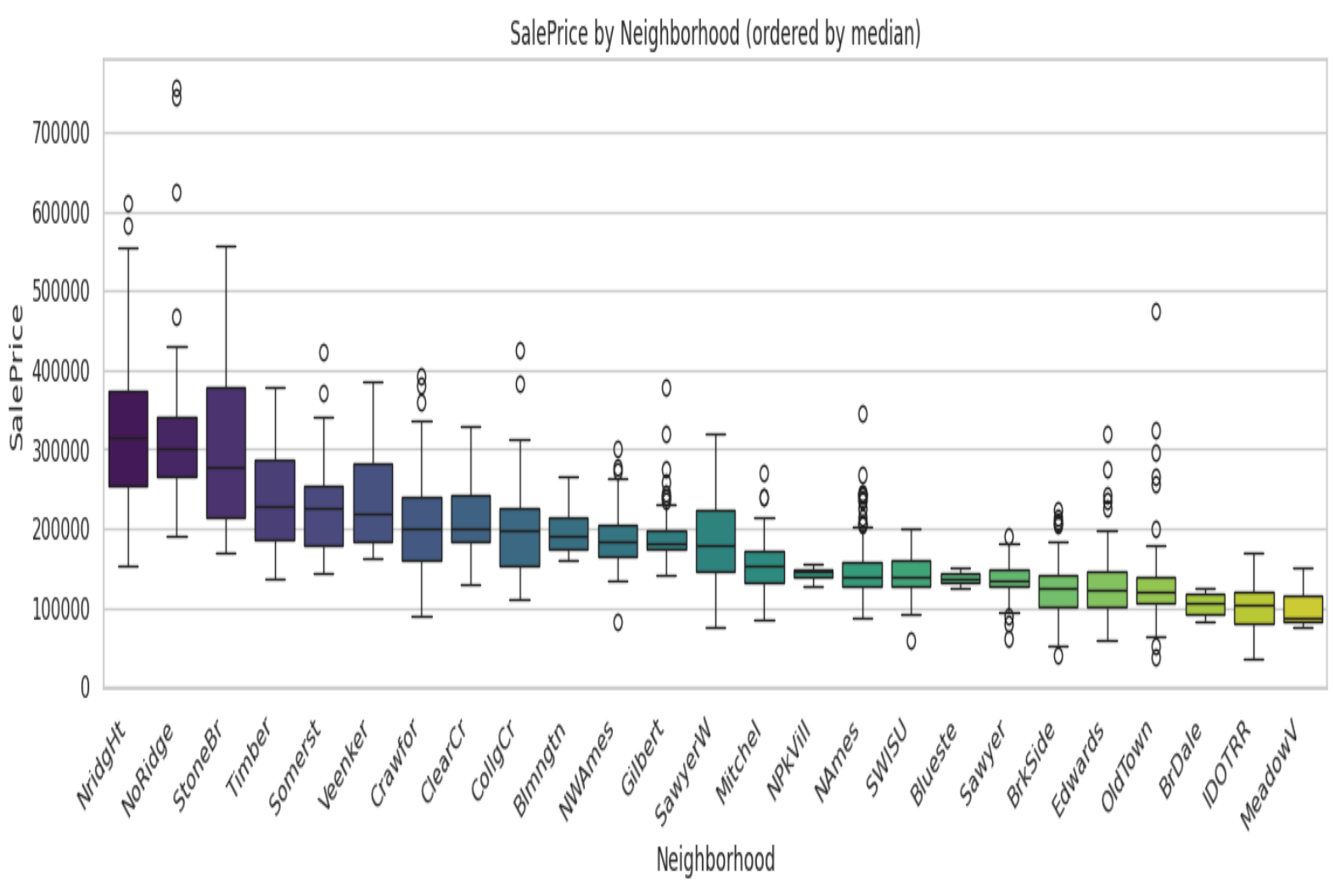


Quality and living area dominate — correlation with SalePrice





Neighborhood premiums and construction era both matter after quality is controlled



INSIGHTS

Top neighborhoods (NoRidge, NridgHt, StoneBr) command 2× the median of lower-tier areas.

Newer construction and recent remodels show clear price premiums. Mild seasonality by month sold is secondary to quality and size.

Engineered HouseAge and RemodAge capture these effects more cleanly than raw year fields.

Domain-aware transforms that preserve information and avoid leakage

01

Impute

“None”/0 for absence features (Pool, Alley, Garage*, Bsmt*). LotFrontage by Neighborhood median. Mode for sparse categoricals.

02

Derive

TotalSF, TotalBath, TotalPorchSF, HouseAge, RemodAge, GarageAge, OverallGrade, binary flags (HasPool, HasFireplace…).

03

Encode

Ordinal maps (Ex→5 … Po→1) for quality/condition. One-hot for remaining nominals. CentralAir → binary.

04

Transform

log1p on continuous features with |skew| > 0.75. Drop near-constant columns (Utilities, Street).

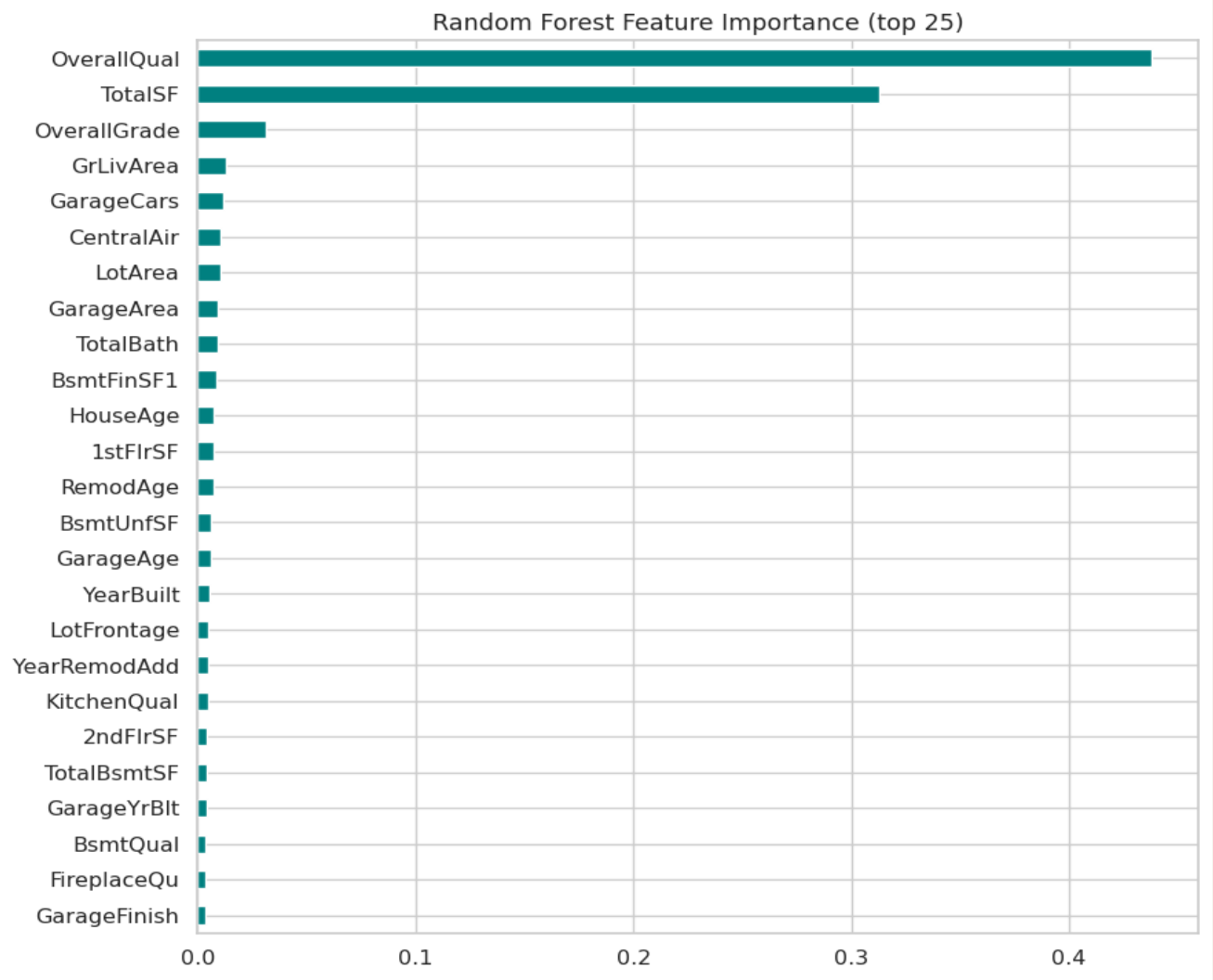
OUTPUT

212 features

After engineering, before selection. All steps fit on train only and applied identically to test.

log1p(SalePrice) used as modeling target to stabilize variance.

212 → 80 features with no loss in cross-validated accuracy



Selection path

212 → ~200

Variance threshold (0.01)

~200 → ~190

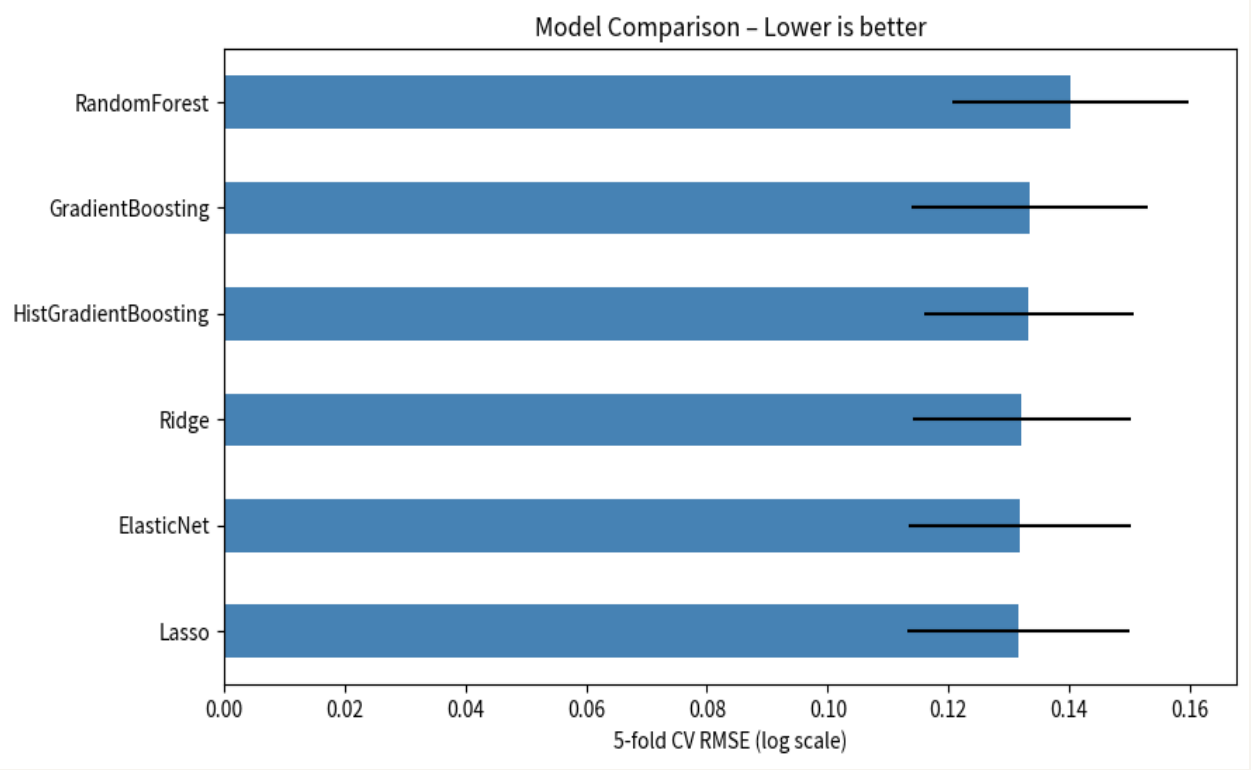
Drop |corr| > 0.95 pairs

~190 → 80

RFE (Random Forest, k=80)

CV RMSE unchanged after reduction — selected set is sufficient and simpler to deploy.

Four models + simple average — lowest OOF error without added complexity



Model	CV RMSE (log)	Std
Ridge (α=50)	0.132	0.018
HistGradientBoosting	0.133	0.017
GradientBoosting	0.134	0.019
RandomForest	0.140	0.019
Ensemble average	**0.129**	—

$15,200

OOF MAE ($)

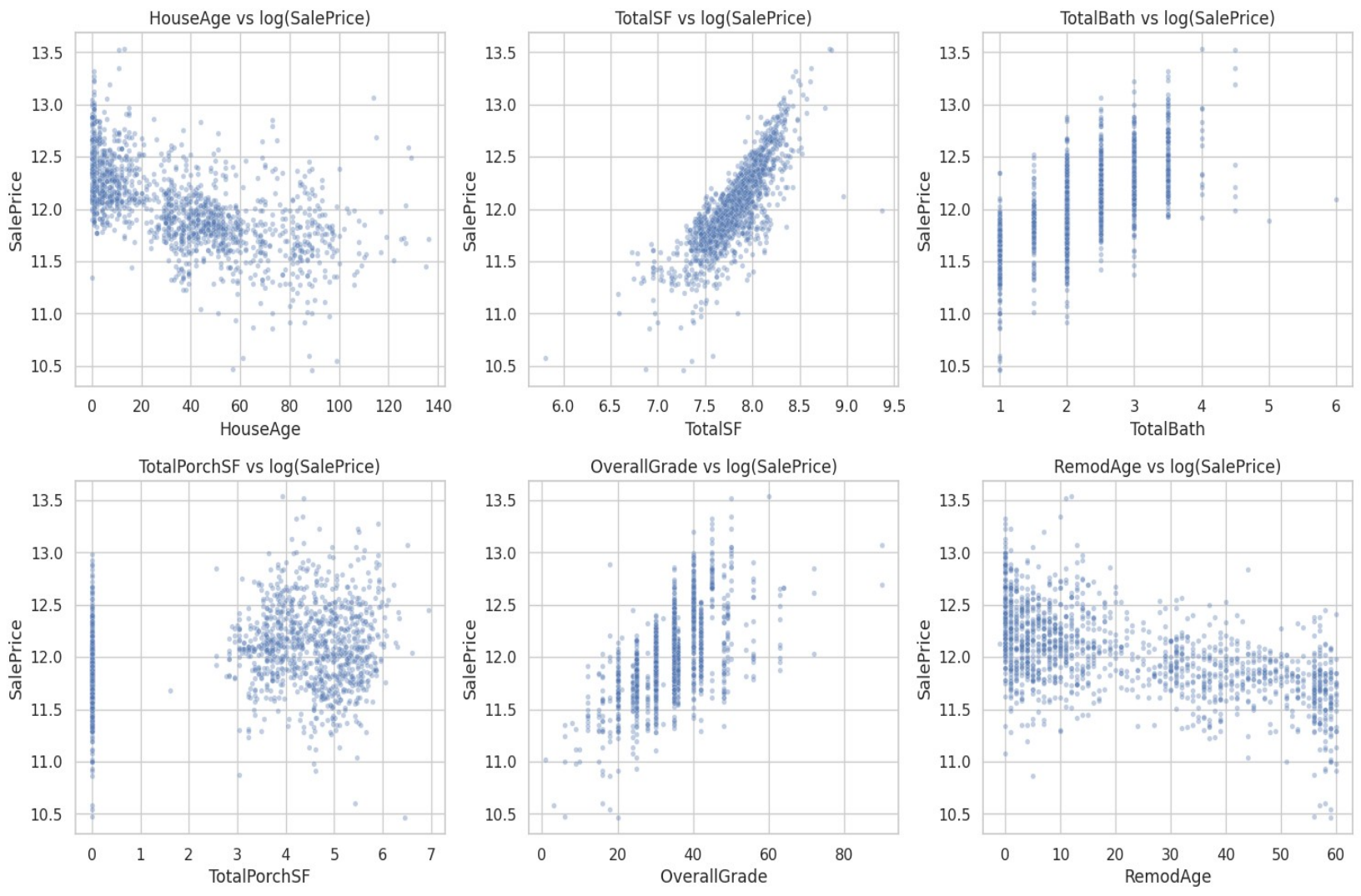
$29,250

OOF RMSE ($)

0.864

OOF R^2

Post-FE relationships confirm TotalSF, age, and quality composites carry signal



WHY THESE FEATURES?

TotalSF aggregates basement + floors into one size measure that models prefer over separate area fields.

HouseAge / RemodAge capture depreciation and update premiums more cleanly than YearBuilt alone.

Binary flags (HasFireplace, HasGarage) give tree models simple high-signal splits.

1,459 predictions — distribution aligns closely with training SalePrice

1,459	$177k	$156k	$54k	$530k
Predictions	Mean pred.	Median pred.	Min	Max

Train vs predicted distribution

Metric	Train	Predicted
Mean	$180,921	$177,154
Median	$163,000	$155,887
Std	$79,443	$73,840
25th pct	$129,975	$127,909
75th pct	$214,000	$207,602

DELIVERY PACKAGE

- submission.csv — Id + SalePrice
- models/ — 4 joblib models
- processed/ — 80-feature matrices
- Kernels/01–06 — audit notebooks
- This management deck (PPT + PDF)

All paths under artifacts/ for full reproducibility.

Deploy · Monitor · Improve

01

Deploy the ensemble

Use the 4-model average for batch valuation. Expected error band $15k–$30k on typical homes. Owner: Analytics. Timing: immediate after approval.

02

Instrument monitoring

Track residual mean and RMSE on new closed sales quarterly. Retrain trigger: RMSE drift > 15%. Owner: Data Eng + Analytics.

03

Document the pipeline

Promote imputation rules, ordinal maps, and the 80-feature list to a versioned production transform. Owner: Analytics. Timing: this sprint.

04

Next-wave improvements

Add LightGBM/XGBoost + stacking; nested target-encoding for Neighborhood; collect post-2010 sales for market-trend features. Expected gain: 2–4% RMSE.

READY FOR DECISION

Approve deployment of the Ames housing price ensemble

OOF RMSE ≈ $29k · R^2 0.86 · 80 features · Full audit trail

Questions & discussion

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